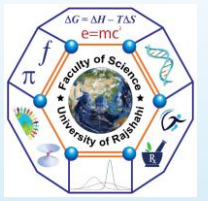




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Measuring Attitudes toward AI's Impact on the Job Market in Bangladesh: A Study at Pabna University of Science and Technology

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Outline

- **Motivation**
- **Research Objectives**
- **Methodology**
- **Key Findings**
- **Conclusion**
- **Future Work**
- **References**

Motivation

- **Real-world need:** AI is reshaping work globally; in Bangladesh adoption is uneven, creating uncertainty about jobs, roles, and skills.
- **Evidence gap at PUST:** No validated campus baseline by role and discipline.
- **Our response:** Build a local baseline on five constructs: AI familiarity, belief in job displacement, job concern, adaptation confidence, willingness to take training.
- **Approach (brief):** Stratified sample $n = 370$, $N = 4,795$; 33-item instrument ($\alpha = 0.813$); KMO = 0.83; nine factors; (57.5%); descriptives, correspondence analysis, ordinal logistic regression.
- **Why it matters:** Target AI-literacy and reskilling, support higher-concern groups, inform curriculum and policy, and track change over time.

Research Objectives

- Measure campus attitudes across five constructs: AI familiarity, belief in job displacement, job concern, adaptation confidence, and willingness to take training.
- Examine how AI familiarity relates to displacement belief, job concern, and adaptation confidence.
- Identify factors linked to higher job concern and adaptation confidence.
- Test whether adaptation confidence increases willingness to take training.
- Establish a local baseline to guide curriculum updates, AI-literacy, and targeted reskilling.

Methodology



Design: Quantitative, cross-sectional survey at PUST



Population & sample: Students, faculty, and staff; **N=4,795**; stratified random; **n = 370**



Instrument: 33-item 5-point Likert on five constructs (AI familiarity, belief in job displacement, job concern, adaptation confidence, and willingness to take training)



Reliability & validity: Pilot tested; Cronbach's $\alpha = 0.813$; KMO = **0.83**; Bartlett's $\chi^2(528) = 3371.182$, $p < .001$

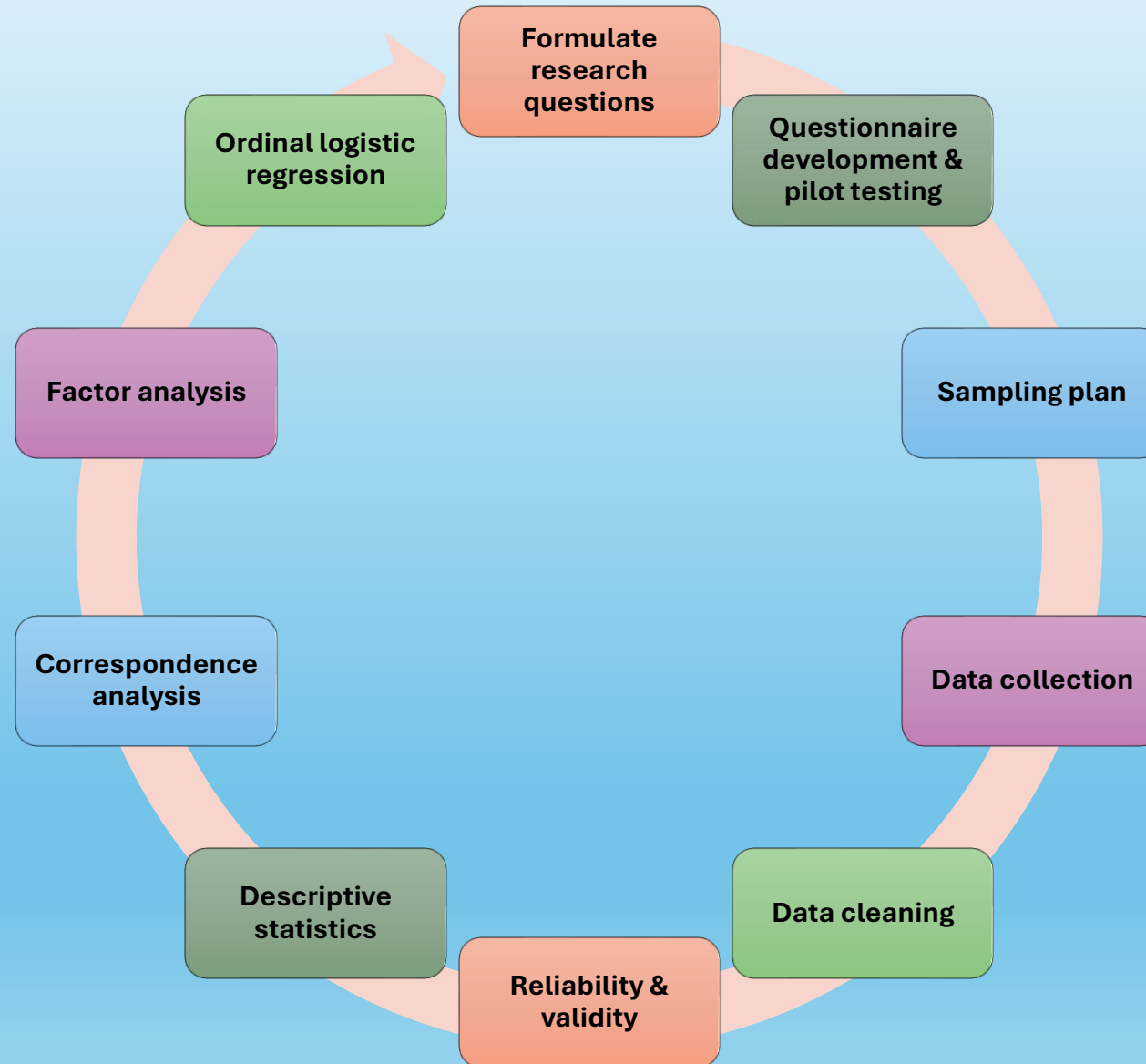


Analysis: Descriptive; correspondence analysis; factor analysis (**9** factors; **57.5%**); ordinal logistic regression on ordered outcomes



Software: IBM SPSS; Microsoft Excel

Flowchart of the methodology



Measurement quality (Reliability)

Scale	Items	Cronbach's α	Interpretation
Overall questionnaire	33	0.813	Good internal consistency

Key Findings

Demographic Insights

Characteristic	Description
Total sample (n = 370)	370 respondents
Age	70.8% aged 15-24
Gender	74.1% male, 25.9% female
Residential status	55.7% urban, 33.0% rural, 11.3% suburban
Roles	343 students, 16 faculty, 11 staff
Response rate	~7.7% of PUST population (N = 4795)

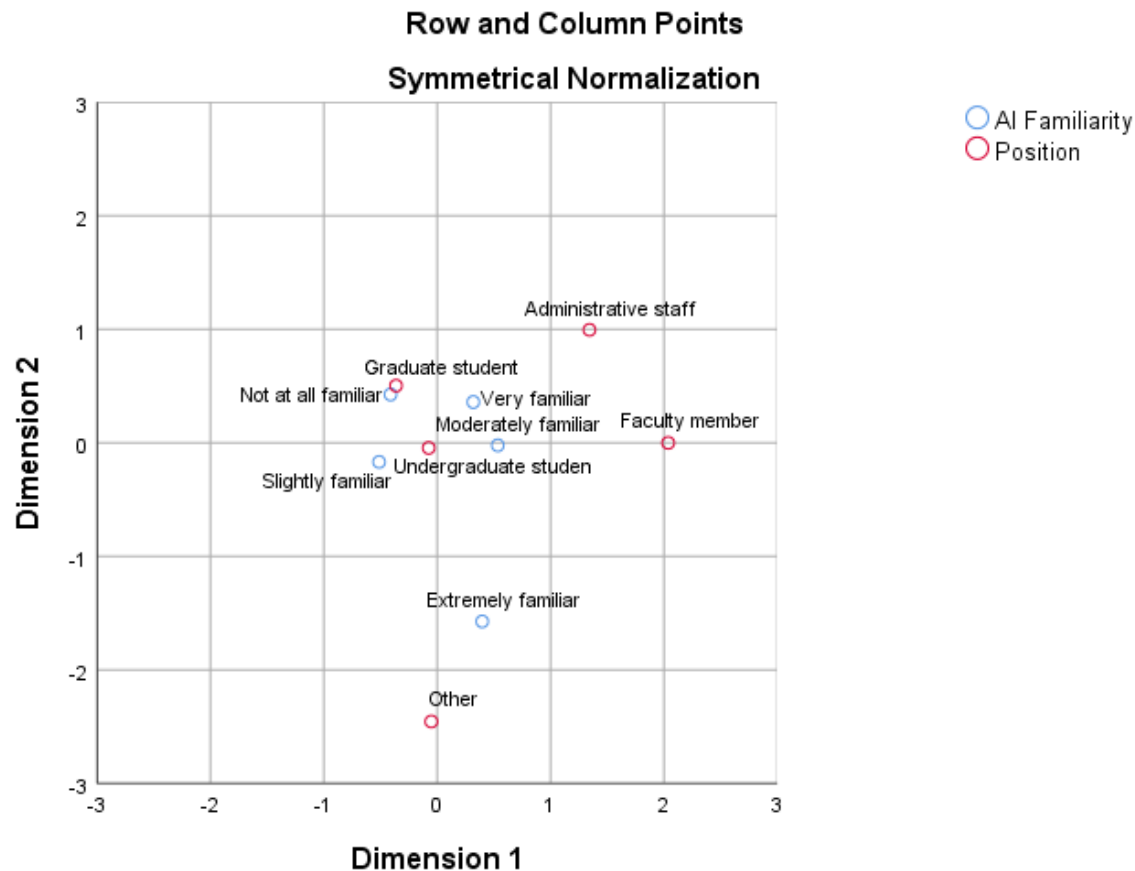
Key Findings on the 5 Constructs

Construct	Key Findings
AI Familiarity	34.6% of respondents are slightly familiar, 30.8% are moderately familiar, 21.3% are very familiar, and 13.8% have no familiarity with AI.
Belief in Job Displacement	52.2% of respondents believe AI could displace jobs, while 27.6% are not concerned about job displacement.
Job Concern	31.9% are slightly concerned, 17.0% are very concerned, and 4.9% are extremely concerned about AI's impact on jobs.
Adaptation Confidence	28.4% are very confident, 32.7% are moderately confident, and 8.4% are not confident at all about adapting to AI.
Willingness to Train	64.3% are likely or very likely to take AI-related training, while 7.3% are unlikely, and 2.7% are very unlikely to pursue training.

Key Findings on Associations (Chi-Square tests)

Association	df	χ^2	p-value (2-sided)	Decision
Belief in job displacement x Gender	4	7.355	0.118	Not significant
AI familiarity x Position	16	27.439	0.037	Significant
Job concern x Age group	8	20.436	0.009	Significant
Adaptation confidence x Work experience	20	23.431	0.268	Not significant
Belief in job displacement x Field of study	16	25.849	0.056	Significant at 0.10 (marginal)
Willingness to take training x Employment status	4	6.300	0.178	Not significant
Job confidence x Income (annual)	16	50.013	<0.001	Significant

AI Familiarity x Position – Correspondence Analysis



AI familiarity varies by position and is highest around faculty, consistent with the chi-square result ($p=0.037$).

Undergraduate sit near the centroid due to large mass; Graduate skew lower. Administrative staff and “Other” are low-mass; interpret cautiously.

Key Findings on Factor Analysis

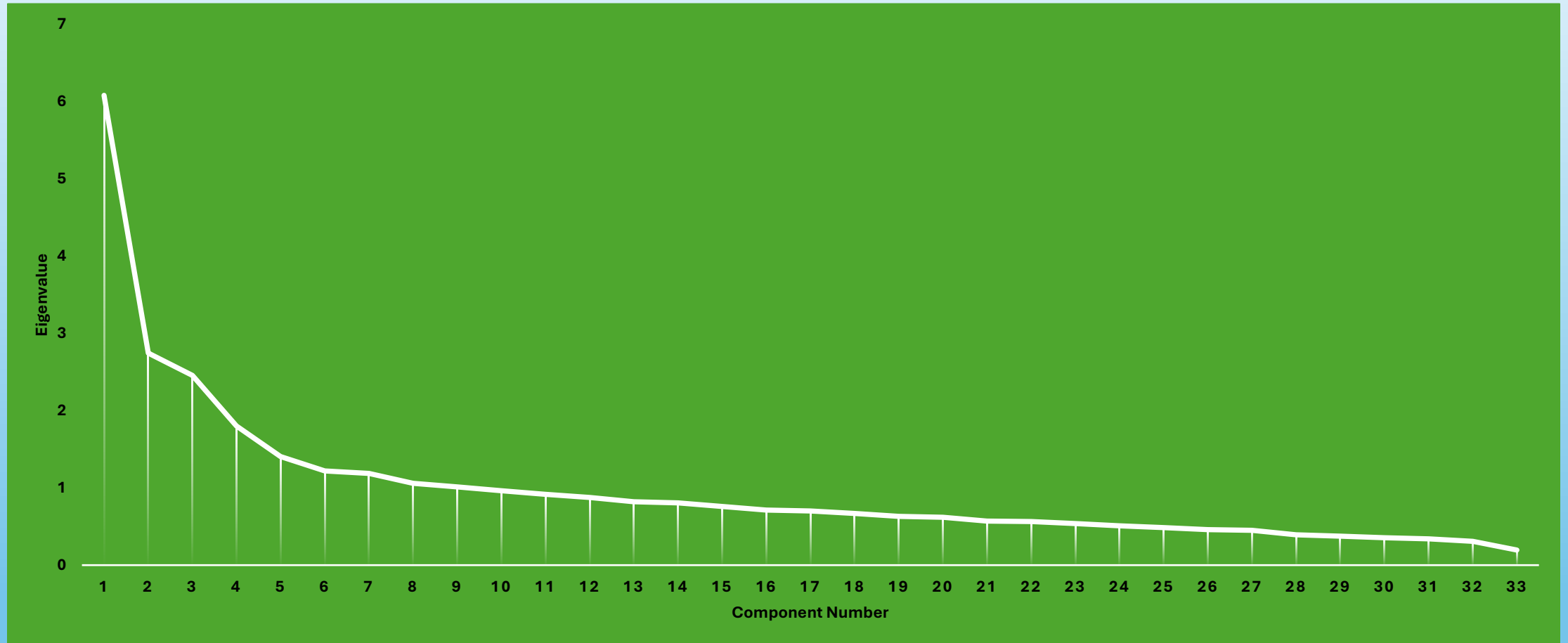
Kaiser–Meyer–Olkin (KMO) & Bartlett’s test of sphericity

Metric	Value	Interpretation
Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy	0.83	Good sampling adequacy
Bartlett’s test of sphericity	$\chi^2(528) = 3371.182, p < .001$	Reject $H_0 \rightarrow$ correlation matrix $\neq$ identity $\rightarrow$ Factor analysis appropriate

Total Variance Explained

Factor	Rotated % of variance	Cumulative %
1	10.593	10.593
2	8.987	19.58
3	8.199	27.779
4	7.074	34.854
5	6.744	41.598
6	4.861	46.459
7	4.043	50.502
8	3.532	54.034
9	3.451	57.486

Scree Plot



Elbow after Factor 9; Kaiser (>1) and scree criteria both support retaining nine factors.

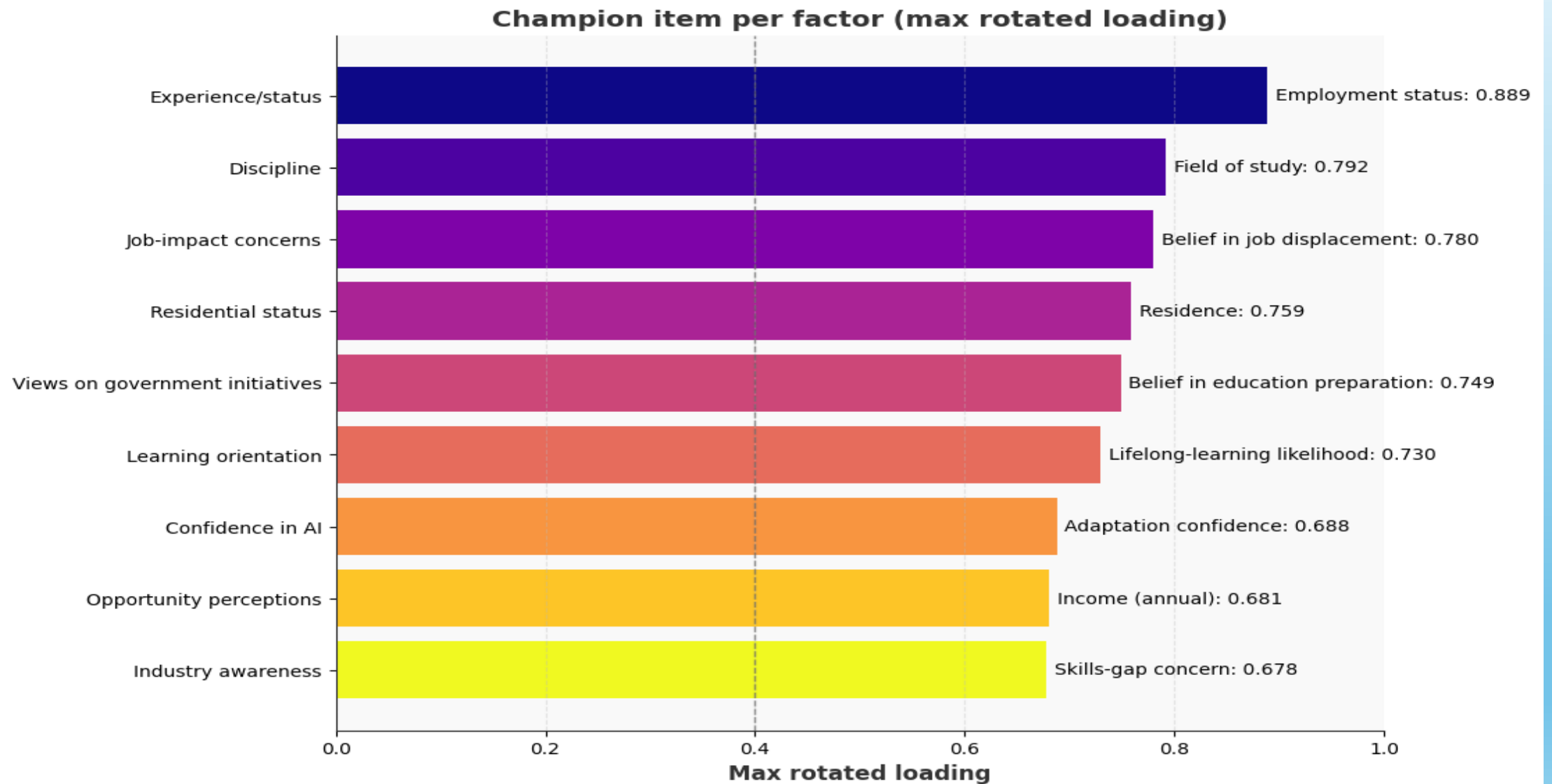
Common Factors

Factor	Label	What it captures	Champion item (max loading)
F1	Learning orientation	Readiness to learn and reskill	Lifelong-learning likelihood (0.730)
F2	Industry awareness	Awareness of AI adoption across sectors	Skills-gap concern (0.678)
F3	Experience/status	Age, role, employment, experience profile	Employment status (0.889)
F4	Confidence in AI	Self-efficacy for using/adapting AI	Adaptation confidence (0.688)
F5	Views on government initiatives	Views on policy, regulation, education readiness	Belief in education preparation (0.749)
F6	Job-impact concerns	Perceived risk of job change/displacement	Belief in job displacement (0.780)
F7	Opportunity perceptions	Expected benefits (productivity, new roles, growth)	Income (annual) (0.681)
F8	Discipline	Field-of-study context	Field of study (0.792)
F9	Residential status	Urban/rural/suburban context	Residence (0.759)

Key Findings on Regression (Ordinal logistic)

Relationship (DV $\leftarrow$ IV)	-2 Log Likelihood	χ^2	df	p-value	Decision and fit
Job concern $\leftarrow$ Age group	56.371	10.314	2	0.006	Significant; good fit
Job concern $\leftarrow$ Work experience	80.640	10.007	5	0.075	Not significant at 0.05; moderate fit (marginal at 0.10)
Belief in job displacement $\leftarrow$ Gender	42.133	1.473	1	0.225	Not significant; poor fit
Adaptation confidence $\leftarrow$ AI familiarity	82.047	56.645	4	<0.001	Significant; good fit
Willingness to take training $\leftarrow$ Adaptation confidence	79.724	62.197	4	<0.001	Significant; good fit

Champion Item per Factor (Max Rotated Loading)



Conclusion

- ❑ **Data suitability:** KMO **0.83**; Bartlett $\chi^2(528)=3371.182$, $p<.001$.
- ❑ **Structure:** **9** factors retained; rotated cumulative variance **57.5%**.
- ❑ **Associations (χ^2):** AI familiarity $\times$ position ($p=.037$); job concern $\times$ age ($p=.009$); job confidence $\times$ income ($p<.001$); belief in job displacement $\times$ field of study ($p=.056$, significant at **0.10**).
- ❑ **Ordinal logistic:** AI familiarity $\rightarrow$ adaptation confidence ($p<.001$); adaptation confidence $\rightarrow$ willingness to take training ($p<.001$); age $\rightarrow$ job concern ($p=.006$); work experience $\rightarrow$ job concern ($p=.075$, not sig. at 0.05).
- ❑ **Cohort signal:** **52.2%** believe AI may displace jobs; **64.3%** are willing to train.
- ❑ **Limitations:** single-institution sample; response rate $\approx 7.7\%$; cross-sectional, self-report; potential non-response bias; ordinal measures; no multi-group invariance/Confirmatory Factor Analysis (CFA) yet.

Future Work

- ❖ **Broaden sample:** multi-campus, public/private, students/faculty/staff.
- ❖ **Track over time:** longitudinal follow-ups on familiarity, confidence, training uptake, and job outcomes.
- ❖ **Strengthen measures:** add objective skill tasks and usage metrics; run CFA and test invariance (gender/discipline).
- ❖ **Extend models:** test mediation (familiarity → adaptation confidence → training) and moderation (age, discipline, residence).
- ❖ **Evaluate programs:** pilot micro-courses; estimate impact with pre/post or randomized/stepped-wedge designs; target the 13.8% with no AI familiarity.

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Thank
You!

Q&A



The End!